

Simplifying algebraic fractions

The fraction $\frac{16}{24}$ can be simplified by dividing the numerator and denominator by 8, as shown.

$$\begin{array}{c} 2 \\ 3 \end{array} \frac{\cancel{16}}{\cancel{24}} = \frac{2}{3}$$

Similarly, the algebraic fraction $\frac{6cd}{2d}$ can be simplified by dividing the numerator and denominator by any common factors.

Thus,

$$\frac{6CD}{2D} = \frac{\begin{array}{c} 3 \\ 1 \end{array} \cancel{6} \times C \times \cancel{D}^1}{\begin{array}{c} 1 \\ 2 \end{array} \times \cancel{D}^1} = \frac{3 \times C}{1} = 3 \times C = 3C$$

Simplify each of the following:

$$\frac{9B^2E}{3B}$$

$$\frac{9B^2E}{3B} = \frac{\begin{array}{c} 3 \\ 1 \end{array} \cancel{9} \times \cancel{B}^1 \times B \times E}{\begin{array}{c} 1 \\ 3 \end{array} \times \cancel{B}^1} = 3BE$$

$$\frac{12a^4b}{ab^2}$$

$$\frac{12a^4b}{ab^2} = \frac{12 \times \begin{array}{c} 1 \\ 1 \end{array} \cancel{a} \times a \times a \times a \times \cancel{b}^1}{\begin{array}{c} 1 \\ 1 \end{array} \times \cancel{a} \times \cancel{b}^1} = \frac{12a^3}{b}$$

1. Using factors to simplify algebraic fractions

The expression $\frac{x^2-4}{x+2}$ can be simplified by first factorizing the numerator and then dividing above and below by the highest common factor.

$$\frac{x^2 - 4}{x + 2} = \frac{(x \cancel{+ 2})(x - 2)}{(x \cancel{+ 2})} = x - 2$$

Simplify the following:

$$\frac{5bx}{20b + 15ab^2} = \frac{5bx}{(5b)(4 + 3ab)} = \frac{x}{4 + 3ab}$$

2. Adding and subtracting algebraic fractions

When adding or subtracting algebraic fractions a common denominator must be used

Simplify the following:

$$\frac{6z}{x(x + 4z)} - \frac{3}{2x}$$

So, to subtract both terms, a common denominator needs to be identified. The common denominator for the above terms is $2x(x + 4z)$.

$$\begin{aligned} \frac{(2)6z}{2x(x + 4z)} - \frac{3(x + 4z)}{2x(x + 4z)} \\ = \frac{(2)6z - 3(x + 4z)}{2x(x + 4z)} \end{aligned}$$

$$\begin{aligned}
 &= \frac{12z - 3x - 12z}{2x(x + 4z)} \\
 &= \frac{-3\cancel{x}}{2\cancel{x}(x + 4z)} \\
 &= \frac{-3}{2(x + 4z)}
 \end{aligned}$$

3. Multiplying algebraic fractions

Algebraic fractions are multiplied by multiplying the numerator and denominators separately.

Simplify the following:

$$\begin{aligned}
 &\frac{2}{x+1} \times \frac{2x}{2x+3} \\
 &= \frac{4x}{(x+1)(2x+3)}
 \end{aligned}$$

4. Dividing algebraic fractions

Algebraic fractions are divided by multiply by the denominator inverted.

Simplify the following:

$$\begin{aligned}
 &\frac{\frac{2}{A+1}}{\frac{2A}{2A+3}} \\
 &= \frac{2}{A+1} \times \frac{2A+3}{2A} \\
 &= \frac{2(2A+3)}{2A(A+1)} \\
 &= \frac{2A+3}{A(A+1)}
 \end{aligned}$$

So, in general to simplify algebraic fractions:

1. A common denominator is needed to add or subtract algebraic fractions.
2. Fractions can be simplified, only if, the numerator and denominator have a common factor.
3. If the denominator or numerator contain fractions added or subtracted, they must be reduced into a single fraction first before proceeding.
4. To divide fractions, multiply by the denominator inverted.